SANTACRUZ ELECTRONICS EXPORT PROMOTION ZONE - AN OVERVIEW



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DEVELOPMENT COMMISSIONER

Ever since the "big Bang" coinciding with Asiad, 1982, the electronics industry in India has expanded at a phenomenal average rate of nearly 35 per cent in output, surpassing the growth in any other industry. The total production of electronic goods (components and equipment) has risen from Rs 8560 million in 1981 to Rs 63,000 million in 1988. By the year 1990-91, the industry is projected to achieve an ambitious target of 108,600 million worth of annual production and this figure is further projected to escalate to Rs 1,000 billion by the end of the century.

Electronics exports have significantly increased from Rs 1145 million in 1983 to Rs 4570 million in 1988. Electronics exports by the end of the Seventh Five Year Plan are targeted at Rs 10,000 million of which the share of software exports is placed at 30%, or Rs 3,000 million. Major factors for the boost in exports have been incentives and liberalised polices and the setting up of a number of Electronic Export Processing Zones. The processing zones now account for over 42% of the total electronics exports from India.

Table 1: Composition of electronics exports

(Value : Rs. million)

Sector	1981	1983	1985	1987
Consumer Electronics	10.0	30.0	50.0	195.0
Communication & broadcasting equipment	19.5	20.0	5.0	20.0
Aerospace & defence equipment	95.5	65.0		
Instrumentation & industrial electronics	29.0	24.0	50.0	417.0
Computers (hardware)	1.0		7.0	48.0
Computers (software)	44.0	170.0	280.0	580.0
Components	80.0	86.0	143.0	560.0
Export processing zones (EPZ)	255.0	750.0	1010.0	1300.0
Total:	564.0	1145.0	1545.0	3120.0

The incentives available to exporters include cash compensatory support, duty drawback and import replenishment licenses. The liberal policy framework includes measures for licensing, import of

Table 2

Share of SEEPZ exports in the country's total electronics exports

Calendar Year	Country's exports (Rs. In million)	SEEPZ exports (Rs. in millions)	Percentage of SEEPZ exports to country's exports
1979	466	116	24.8
1980	418	165	39.4
1981	564	255	45.2
1982	839	483	57.5
1983	1,145	750	65.5
1984	1,550	1,035	66.5
1985	1,545	850	55.0
1986	2,400	1,005	42.0
1987	3,120	1,097	35.0
1988	4,570	1,420	31.0

Product Groupwise Exports

Table 3

Product groupwise percentage in the total exports from SEEPZ

CATEGORY	Percentage during					
	1984-85	85-86	86-87	87-88	88-89	89-90
Electronics :						
a) Computers, computer sub-assemblies and peripherals etc.	64	57	58	62	54	48
b) Other electronic components	26	24	19	18	11	8
c) Consumer electronics	5	13	13	8	18	8
d) Computer software	4	5	8	11	9	7
e) Instruments and communication equipment	1	1	2	1	1	1

technology, and investment by foreign companies. Duties and taxes on capital; goods, raw materials, components and finished products have been rationalised in order to enable manufactures' production at economies of scale, at near international prices and based on comtemporary technology.

An exclusive Electronics Export Promotion Zone has been set up at Santacruz in Bombay. In addition there are five other export processing zones located at Kandla (Gujarat), Madras (Tamil Nadu), Falka (West Bengal), Noida (U.P.) and Cochin (Kerala). Export production has originated both from the units in the export processing zones and those in the Domestic Tariff Areas (DTA)

Table 4

INCENTIVES AND CONCESSIONS

1. SEEPZ units are entitled to a tax holiday for any five consecutive years of the first eight years from the commencement of their production.
2. Total waiving of licensing for import of capital goods and other production materials. An Open General Licence System is force which provides for the import of all goods without and licence.
3. Duty-free import to capital goods and equipment from preferred sources.
4. Exemption from customs duty on imported raw materials, Components, consumables, spares, tooling and packaging materials.
5. Exemption from Central Excise duties and other levies on products manufactured within the Zone.

The Santacruz Electronics Export Promotion Zone (SEEPZ) set up by the Government of India in Bombay in 1974 had in the eighties contribution to the extent of more than 66 per cent of the total electronics exports from the country. How well time and judicious was that decision is amply borne out by the broadening range, increasing volume and the internationally accepted quality of SEEPZ exports during the past few years.

The Sector-wise break up of electronics exports during the period 1981-1987 is given in Table-1, the share of SEEPZ exports out of the country total electronics exports in Table-2 and the product-groupwise percentage in total exports from SEEPZ during the last five years in Table 3. Most portion of the exports here comprised to floppy disk drives, wincheter heads and other computer peripherals in high technology areas.

Both the Central and State Governments have extended various facilities and concessions to the entrepreneurs who have established production facilities. A list of the incentives and concessions offered is listed in Table 4.

The zone provides plots to entrepreneurs desiring to put up their own factory premises to suit their specific requirements. It offers ready built space in the standard Design factory Buildings of which there are at present five, each with 2,00,000 sq.ft of built up area. In order to enable SEEPZ computer software units to transmit software through satellite link, data transmission facilities have also been commissioned at SEEPZ by Videsh Sanchar Nigam limited (VSNL) and Mahanagar Telephone Nigam (MTNL) authorities.

Right from its inception, the SEEPZ administration has never

Table 5 Foreign Collaborations	
	Financial Technical & Marketing
USA	26
UK	2
Denmark	1
Belgium	1
HongKong	5
Japan	4
West Germany	3
Gulf countries	1

compromised on the character and the high standards of infrastructure, so much so that it has come to be recognised as a model for emulation by others. The administration has also been most helpful and co-operative towards the entrepreneurs and has extended all possible assistance in ensuring maximum exports. But in an item like electronics, the time factor is important whether it relates to scheduled deliver or imports of raw materials or capital goods. The SEEPZ manufactures' association is at present attending to these problems (See Box)

Some of the more enterprising electronics units in the zone have won such ready acceptance of their products in the developed countries such as the United States, Japan and West Germany that they have appropriated a tidy slice of the international market. Their products not only measure up to world standards but even go one step further to defy competition from their counterparts in the developed countries.

Of the 8 electronics units in the zone, giving employment to a total of under 10,000 personnel, 43 have foreign collaboration agreements. The number of units fully owned by foreign parties is about 10, and the total foreign equity investments in the units set up in the zone amount to about Rs 700 crores. A list of country-wise foreign collaborations is given in Table 5, and a destination wise percentage in the total exports form SEEPZ during the last five years in Table 6.

Software exports in another activity SEEPZ witnessing a remarkable growth. The exports of software amounted to Rs 27 crores in 1989-90. India is already recognised worldwide for software as the Far East for computer hardware. The exports for software have received a particular boost after the liberalised policy for computer software was announced in December 1986.

Over the last couple of years, multinations have started setting up their offshore development centers in SEEPZ. Success of this activity has already been demonstrated by pioneers in this business like Tata Unisys Ltd. (TUL), a joint venture established in 1977 between Tata-Sons and Unisys Corporation of the United States, each of them holding 40% equity and the remaining 20 being with the public. Companies producing software in the zone at present include Tats Unisys, Mahendra BT, Unique Data software, Digitron, Wang Computers, Mastek, etc.

As it is, SEEPZ has emerged as a pattern of a Free Trade Zone for emulation by others existing elsewhere in the country. The percentage fall in electronics exports from SEEPZ in the last couple of years is because of the increasing share of exports from DTA and other FTZS. Significantly, a Gem, & Jewellery Export Complex has been added to the zone in 1986-87 to boost exports in this commodity. Already these exports account for 28 per cent of SEEPZ's total exports in 1989-90, just two years after its inception.

Over the years, SEEPZ has been functioning as a nursery which has nurtured the growth of a number of electronics units. Many such units have forged a head to carve out a niche for themselves in the international market. In what follows, more detailed information on some such units are given.

Table 6 Destination wise exports						
	Percentage					
COUNTRY/AREA	1984-85	85-86	86-87	87-88	88-89	89-90
a) Far East	48	40	40	48	27	30
b) USA/Canada	33	38	30	26	26	32
c) East Europe/ USSR (Rupee payment exports)	7	13	20	15	36	21
Hard currency exports	—	—	—	1	—	—
d) West Europe & UK	9	7	8	8	10	16
e) Other Countries	2	2	2	2	1	1

lishment of the export subsidiary in SEEPZ was two fold.

(i) To project the capability of Indian Engineers to produce high technology products at international quality and prices, and

(ii) To discuss benefits from the company's exposure to the stringent requirements of the world market and apply the technical innovations to products made for the domestic market.

After successfully marketing the LAB series and compact Series Regulated Power Supplies and Line Conditioning Equipment, the company has introduced a large variety of Power Supply Systems.

In 1989-90 the company bagged a large order for Electronic Exchange from a European company with an international marketing set up. The nineties are expected to bring substantial growth in this business sector. The company's industrial products like Industrial Printers and Video Monitors have also recorded substantial growth in 1990. Having successfully marketed Oscilloscopes, Test Instruments & Power Supply Systems in Europe for the last ten years, Intel Instruments & Systems have identified U.S.A and Canada as thrust areas in the nineties.

SEEMA PLAYING BIG ROLE IN EXPORTS



Mr. Tandon
President

SANTACRUZ ELECTRONIC EXPORT MANUFACTURERS Association was set up in early 1974 and revitalised in 1982. The present Chairman M.L. Tandon, is enthusiastic about free Electronic Export Business. According to him giving licences is not the end. There are plenty of problems, non flexibility in diversification of product manufacturing according to the buyer, Finance, Customs, Communication, Delivery, Power, Labour and so on.

The foreign buyer expects quality prompt delivery, proper prices, and incentives. Now he can get to his terms and conditions in other countries. Why should he then come here. He is not interested in our problems.

With the co-operation of the Department of Electronics, Ministry of Finance and Ministry of commerce, and Seepz Development Commission situation is better. The Association takes keen interest in receiving delegations and helping the units to function well.

New jewellery units have sprang up. Association also extend its co-operation to them.

DIGITAL EQUIPMENT EXPORT (INDIA) LTD

This is a wholly owned subsidiary of Digital Equipment (India) Ltd, and has set up a **DIAGNOSTICS AUTOMATIC TEST EQUIPMENT (ATE) SOFT WARE DEVELOPMENT CENTRE AT SEEPZ.** This facility started operations in 1985 with the aim to provide quality Diagnostics & ATE Software to DEC's customer Service Logistics Group at very cost effective rates and predefined time schedules.

The facility has very widely used ATE like GENRAD's GR 2272, FAIRCHILD's FF333 & AFIT 3000C, and FLUKE's 9010A. DEE(I) L has till now developed more than 200 test programs for DEC, U.S.A. on these testers since its inception, meeting all the schedules, and

TANDON INDIA - EXCELLENCE IN ELECTRONICS EXPORTS



Mr. M.L. Tandon

Tandon-India, a group of Tandon companies for exports and Indian markets occupy 28,000 square metres of space in the SEEPZ area. The group headed by **Mr. M.L. Tandon** as Chairman & Managing Director is a subsidiary of the Tandon Corporation of United States, a world reknowned firm in computer systems.

The Indian companies produce a wide variety of products ranging from Winchester Head Assemblies, Floppy Disk Drives Computer Systems, Stepper Motors, Switch Mode Power Supplies and Electronic sub-assemblies. This has been made possible with the excellent in-house capabilities for machining, plastic injection moulding, PCB assembly and Automatic Test Equipment (ATE) as given in Table 1.

Coil Winding & Surface Treatment Machines

Cable Assemblies & Harness Assemblies

Capabilities of Circuit Designing & Prototype Building

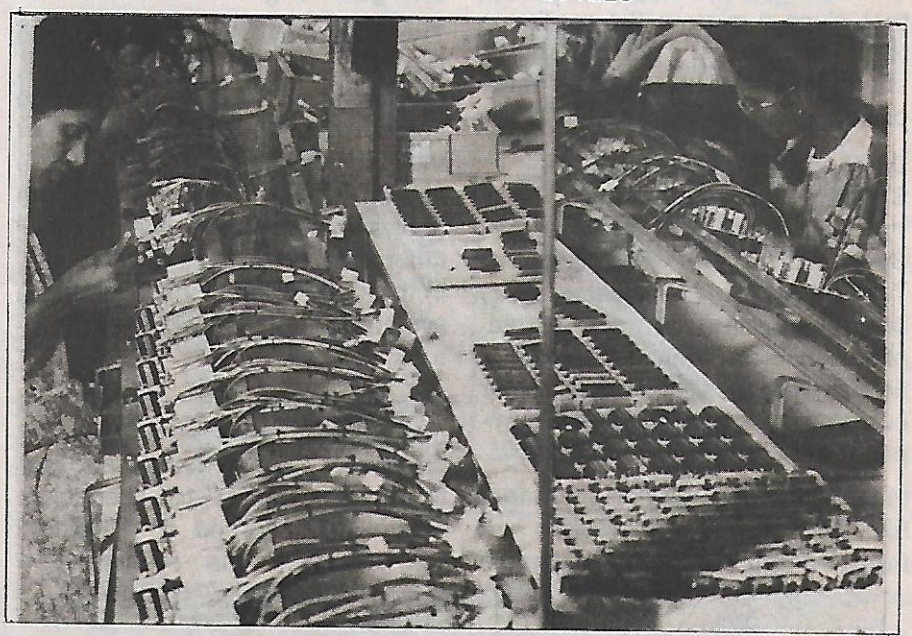
The group consists of seven separate companies for exports and another companies domestic market. The names of these companies along with the products they deal with are given in Table 2 A list of products produced by the companies is given in Table 3 and their cumulative export performance to-date in Table 4. Tandon India has worldwide representative offices and service centres in Singapore (South-east Asia), Los Angeles (USA) and Frankfurt (EUROPE)

TABLE 1
IN-HOUSE CAPABILITIES OF CRITICAL COMPONENTS & PROCESSES

MACHINING	Numerically controlled machines like FANUC, MAHO, MICRON
MOULDING	Plastic Injection Moulding Machines like ARBURG.
PCB ASSEMBLY	Stuffing, Soldering & Cleaning Facility
A.T.E.	In-circuit & Functional Test Systems like Hp-3062

INDIAN MARKETS

MEMORY ELECTRONICS PVT LTD.	Floppy Disk Drives, Computer Systems
TANDON HOLDINGS PVT. LTD.	Sheet Metal & Spring Parts
GOLDEN COMPUTERS LTD.	Sales & Services
TANDON DATA SYSTEM SYSTEM LTD.	Application Softwares
TOTAL SPACE OCCUPIED	28,000 SQ. MTRS
TOTAL NO. OF EMPLOYEES	2000



View of the PCB assembling at the Ultratek's switch mode power supply unit

TABLE 2
TANDON COMPANIES FOR EXPORTS & INDIAN MARKETS

EXPORT

TANDON MAGNETICS (I) PVT. LTD.	Floppy Head Assemblies
TANDON MOTORS PVT. LTD.	Stepper & Spindle Motors
TANCOM ELECTRONICS	Winchester Head Assemblies
ADVANCE TECHNOLOGY DEVICES	Winchester Head Assemblies of the latest designs
MODULER ELECTRONICS (B) PVT. LTD. - at Madras -	
ULTRA TEK DEVICES PVT. LTD.	Switch Mode Power Supplies
EASTERN PERIPHERALS PVT. LTD.	Floppy Disk Drives, Computer Systems

TABLE 3

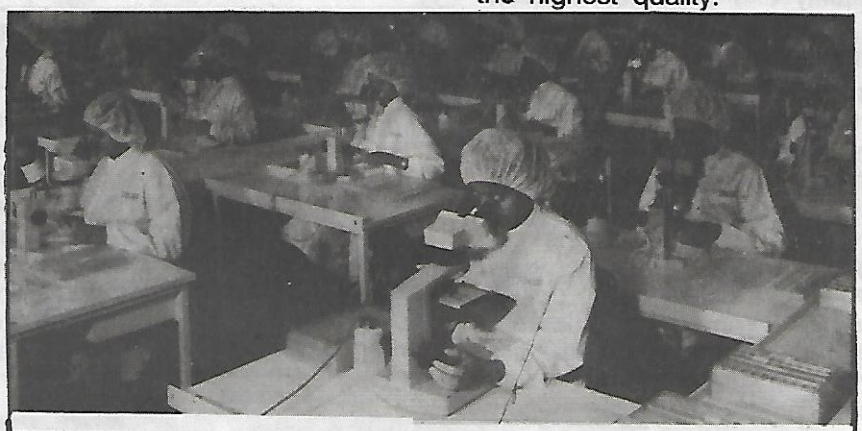
PRODUCT LIST

- SWITCH MODE POWER SUPPLIES
- WINCHESTER HEAD ASSEMBLIES
- FLOPPY DISK DRIVES
- STEPPER MOTORS
- SPINDLE MOTORS
- COMPUTER SYSTEMS
- ELECTRONIC SUB-ASSEMBLIES

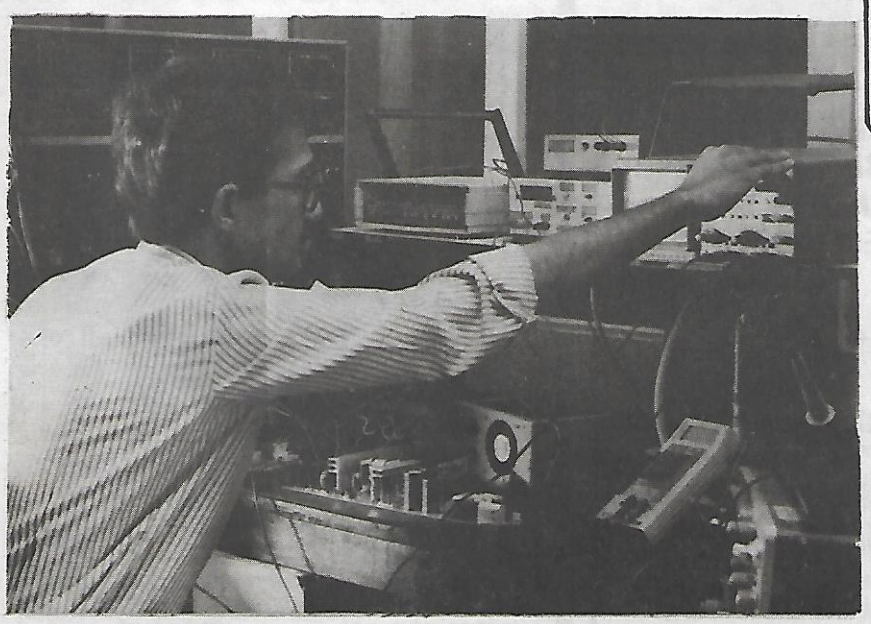
Experience:

The dedication to excellence has carved a niche to Tandon Group of Companies.

Within a egalitarian environment each associate is given an opportunity to pursue individual ambition for the good of group goals. Each individual is given an sense of pursuing a common goal and each has a specific role that contributes to that goal and the goal in creating and marketing Products and providing service of the highest quality.



Tancom Electronics Winchester head assembling in full swing on a mass producing facility



Tandon's R & D's indigenious efforts have made their design acceptable world over.

TABLE 4

CUMULATIVE EXPORT PERFORMANCE

	No. of units
- FLOPPY DISK DRIVES	- 2.05 million
- SWITCH MODE POWER SUPPLIES	- 500 THOUSAND
- STEPPER MOTORS	- 5.05 MILLION
- FLOPPY HEAD ASSEMBLIES	- 14.00 MILLION
- WINCHESTER HEADS	- (HGA) 11 MILLION (HEADS)
- COMPUTER SYSTEMS	- 25000

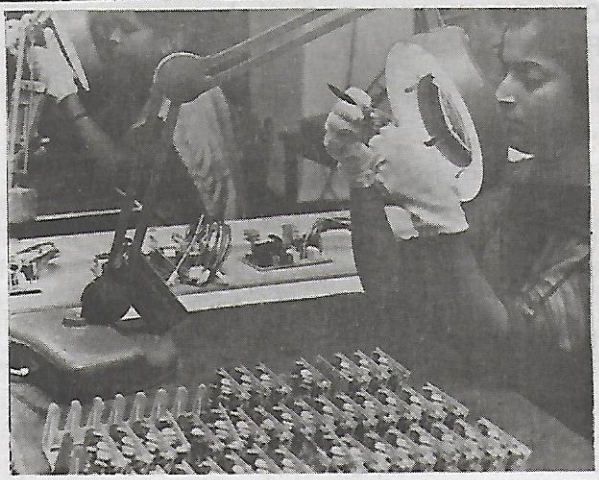
With the judicious blending of hands and automation the Tandon Products have made a name in U.S.A. and western European market.

The product line being the highest state of the art ranging from 8088/80286/80386/80486 computer systems, Tandon data pac is only one of its kind in the world for storage data in a portable/secured pac giving an unbeatable combination.

All the products have been approved by TUO, CSA and UL for safety and EMI requirements. The Products meet the toughest international standards.

Mass Production facility

Covering an area of 28,000 sq mt. every unit of Tandon has a mass manufacturing facility which has a capacity of producing of high scale mass units of computer system, floppy disk drive and switch mode power supplies, Winchester heads, floppy heads, Stepper motors, etc.



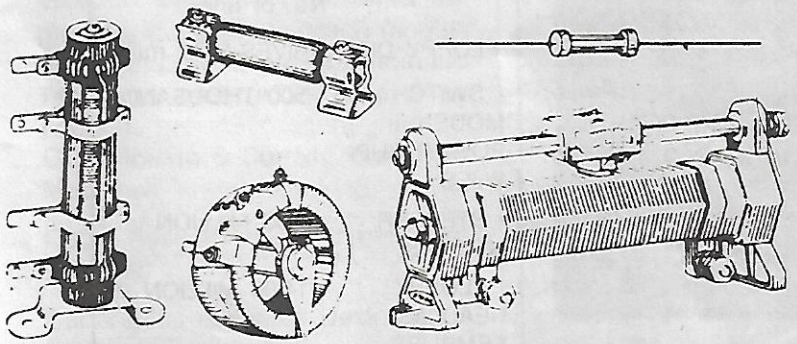
MINUTE AUDITING OF THE P.C.B. UNDER A MAGNIFIER

Tandon's contribution account for 40/50 per cent of total electronics export from Seepz and 30% of India's total electronics export. This is really an achievement and tribute for the ability of Tandon's Products of world standard to compete in the international market.

Tandon India with the help of its excellent resources like management expertise, a dedicated work force of about 2000 Persons appropriate technology and processes and excellent in house manufacturing facilities have indeed became hi-tech manufacturing company caterers to the changing customer and market requirements both at home and abroad.

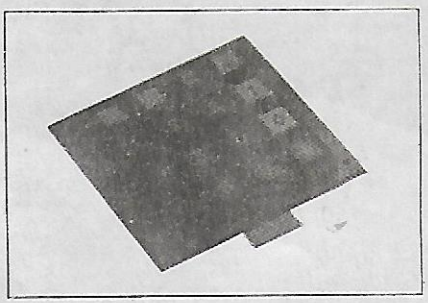
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